Biostatistical Methods II Midterm Cheatsheet (Lectures 1-11)

II. Exponential Family (Lecture 2)

Standard Form:

$$f(y;\theta,\phi) = \exp\left\{\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right\}$$

- θ: Canonical parameter. ϕ: Dispersion parameter.
- **Mean:** $E(Y) = \mu = b'(\theta)$.
- **Variance:** $\text{Var}(Y) = a(\phi)b''(\theta) = a(\phi)V(\mu)$.
- **Variance Function:** $V(\mu) = b''((b')^{-1}(\mu))$.

Common Distributions:

1. **Normal:** $f(y) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp[-\frac{(y-\mu)^2}{2\sigma^2}]$
   - $\theta = \mu, b(\theta) = \theta^2/2, a(\phi) = \sigma^2$.
   - $E(Y) = \mu, \text{Var}(Y) = \sigma^2$.
2. **Poisson:** $f(y) = \frac{e^{-\lambda}\lambda^y}{y!}$
   - $\theta = \log\lambda, b(\theta) = e^\theta = \lambda, a(\phi) = 1$.
   - $E(Y) = \lambda, \text{Var}(Y) = \lambda, V(\mu) = \mu$.
3. **Binomial ($n = 1$):** $f(y) = \pi^y(1-\pi)^{1-y}$
   - $\theta = \log\frac{\pi}{1-\pi}, b(\theta) = \log(1+e^\theta), a(\phi) = 1$.
   - $E(Y) = \pi, \text{Var}(Y) = \pi(1-\pi), V(\mu) = \mu(1-\mu)$.
4. **Gamma:** $f(y) = \frac{\beta^\alpha}{\Gamma(\alpha)}y^{\alpha-1}\exp(-\beta y)$
   - $\theta = -1/\mu = -\beta/\alpha, b(\theta) = \log\mu = \log(\alpha/\beta), a(\phi) = 1/\alpha$.
   - $E(Y) = \alpha/\beta = \mu, \text{Var}(Y) = \alpha/\beta^2 = \mu^2/\alpha, V(\mu) = \mu^2$.

Proving a distribution is EF: Take the log of the density, use $\exp(\log(\cdot))$, and try to isolate $y \cdot$ (function of θ) as the first term. If y is multiplied by something that depends on θ, that "something" is your θ (if canonical link used).

III. GLM Framework (Lecture 3)

Three Components:

1. **Random:** $Y_i \sim$ EF (Independent).
2. **Systematic:** $\eta_i = \mathbf{x}_i^T\boldsymbol{\beta} = \beta_0 + \beta_1 x_{i1} + \ldots$.
3. **Link Function:** $g(\mu_i) = \eta_i$. (Canonical link: $g(\mu) = (b')^{-1}(\mu) = \theta$).

Estimation (Fisher's Scoring):

- **Score Function:** $U(\boldsymbol{\beta}) = \frac{\partial\ell}{\partial\boldsymbol{\beta}}$, $E[U] = 0$.
- **Fisher Information:** $\mathcal{I}(\boldsymbol{\beta}) = E[-\frac{\partial^2\ell}{\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^T}] = \text{Var}(U)$.
- **Update Rule:** $\boldsymbol{\beta}^{(m)} = \boldsymbol{\beta}^{(m-1)} + \mathcal{I}^{-1}(\boldsymbol{\beta}^{(m-1)})U(\boldsymbol{\beta}^{(m-1)})$.
- **Asymptotic Normality:** $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \mathcal{I}^{-1}(\hat{\boldsymbol{\beta}}))$.

IV. Inference (Lecture 4)

Deviance (D): Measure of discrepancy between fitted and saturated model.

$$D = 2\{\ell(\text{saturated}) - \ell(\text{fitted})\}$$

- **Model Comparison:** For nested models $M_0 \subset M_1$, $D_0 - D_1 \sim \chi^2_{p_1-p_0}$.

Residuals:

1. **Pearson:** $r_{P_i} = \frac{y_i - \hat{\mu}_i}{\sqrt{V(\hat{\mu}_i)}}$. $\sum r^2_{P_i} = X^2$ (Pearson χ^2).
2. **Deviance:** $r_{D_i} = \text{sign}(y_i - \hat{\mu}_i)\sqrt{d_i}$, where $\sum d_i = D$.

Three Hypothesis Tests ($H_0 : \boldsymbol{\beta} = \boldsymbol{\beta}_0$ where $\boldsymbol{\beta}$ has q constraints):

- **Wald:** Uses MLE and its curvature. $TS = (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)^T\hat{\mathcal{I}}(\hat{\boldsymbol{\beta}})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \sim \chi^2_q$.
- **Score:** Uses the slope of the likelihood at H_0. $TS = U(\boldsymbol{\beta}_0)^T\mathcal{I}^{-1}(\boldsymbol{\beta}_0)U(\boldsymbol{\beta}_0) \sim \chi^2_q$.
- **LRT:** Uses the difference in log-likelihood heights. $2\{\ell(\hat{\boldsymbol{\beta}}) - \ell(\boldsymbol{\beta}_0)\} \sim \chi^2_q$.
  - *Note:* For a single parameter $H_0 : \beta_j = \beta_{j0}$, *df* $= 1$. Use $z = \frac{\hat{\beta}_j - \beta_{j0}}{SE(\hat{\beta}_j)} \sim N(0,1)$ (which is $\sqrt{\chi^2_1}$).

V. Binary Outcomes (Lectures 5-6)

Link Functions:

- **Logit (Canonical):** $\log\frac{\pi}{1-\pi} = \eta$.
- **Probit:** $\Phi^{-1}(\pi) = \eta$.
- **Complementary Log-Log:** $\log(-\log(1-\pi)) = \eta$ (Asymmetric, for extreme values).

Interpretation of Logistic Regression:

- β_j is the **change in log-odds** for a 1-unit increase in x_j.
- $OR = \exp(\beta_j)$ is the **Odds Ratio**.
- Probability $\pi = \frac{e^\eta}{1+e^\eta}$.

Confidence Intervals (CI):

- **For** β**:** $\hat{\beta} \pm z_{\alpha/2}SE(\hat{\beta})$.
- **For** π**:** Get CI for $\eta = \mathbf{x}^T\boldsymbol{\beta}$ first! $[L, U] = \hat{\eta} \pm z\sqrt{\mathbf{x}^T\text{Cov}(\hat{\boldsymbol{\beta}})\mathbf{x}}$.
- Then transform: $[\frac{e^L}{1+e^L}, \frac{e^U}{1+e^U}]$.

Goodness of Fit (GOF):

- **Grouped Data (m_i large):** Use D or $X^2 \sim \chi^2_{n-p}$.
  - **Pearson χ^2:** $X^2 = \sum_{i=1}^n \frac{(y_i - m_i\hat{\pi}_i)^2}{m_i\hat{\pi}_i(1-\hat{\pi}_i)}$

- **Deviance:** $D = 2\sum_{i=1}^n \left\{ y_i \log \frac{y_i}{m_i\hat{\pi}_i} + (m_i - y_i)\log\frac{m_i - y_i}{m_i(1-\hat{\pi}_i)} \right\}$
- *Note:* If $y_i = 0$ or $y_i = m_i$, the deviance is still well-defined.

- **Ungrouped Data ($m_i = 1$):** D and X^2 are unreliable. Use **Hosmer-Lemeshow** test.
  - Group into $g \approx 10$ categories. $\chi^2_{HL} = \sum_{j=1}^g \frac{(O_j - n_j\bar{\pi}_j)^2}{n_j\bar{\pi}_j(1-\bar{\pi}_j)} \sim \chi^2_{g-2}$
  - O_j: observed 1's in j-th group (size n_j), $\bar{\pi}_j$: avg estimated prob in j-th group. Large value $\Rightarrow$ lack of fit.

VI. Overdispersion (Lecture 7)

Definition: $\text{Var}(Y) = \phi \cdot \text{Var}_{nominal}(Y)$ where $\phi > 1$.

- **Note:** Overdispersion **cannot** occur in ungrouped binary data ($n_i = 1$).

Diagnostics:

- **Estimation:**
  - Pearson-based: $\hat{\phi} = \frac{X^2}{n-p} = \frac{1}{n-p}\sum_{i=1}^n \frac{(y_i - m_i\hat{\pi}_i)^2}{m_i\hat{\pi}_i(1-\hat{\pi}_i)}$ (Generally preferred).
  - Deviance-based: $\hat{\phi} = \frac{D}{n-p}$.
- **Half-normal plot:** If points fall outside the envelope or the slope is > 1, overdispersion is likely. Empirical slope $\approx \sqrt{\phi}$.

Corrective Actions:

- **Estimates:** $\hat{\beta}_Q = \hat{\beta}_{MLE}$, but $\text{Var}(\hat{\beta}_Q) = \hat{\phi}\text{Var}(\hat{\beta}_{MLE})$.
- **Hypothesis Testing ($H_0 : \beta = \beta_0$):**
  - **Wald/Score Test:** $TS_{adj} = TS/\hat{\phi} \sim \chi^2_q$.
  - **Deviance Analysis (F-test):** For nested models $M_0 \subset M_1$, use:

$$F = \frac{(D_0 - D_1)/(p_1 - p_0)}{\hat{\phi}} = \frac{(D_0 - D_1)/q}{X^2/(n-p_1)} \sim F_{q,n-p_1}$$

where X^2 is from the larger model (M_1).

VII. Binary Topics (Lectures 8-9)

LD50 (Median Lethal Dose): Dose x such that $\pi(x) = 0.5 \Rightarrow \eta = 0$.

- For $\eta = \beta_0 + \beta_1 x$, $\hat{x}_{LD50} = -\hat{\beta}_0/\hat{\beta}_1$.

Delta Method (Variance of LD50):

$$\text{Var}(\hat{x}_{LD50}) \approx \frac{1}{\beta_1^2}\text{Var}(\beta_0) + \frac{\beta_0^2}{\beta_1^4}\text{Var}(\beta_1) - \frac{2\beta_0}{\beta_1^3}\text{Cov}(\beta_0, \beta_1)$$

VIII. Categorical Y (Lectures 10-11)

1. Nominal (Multinomial) Response

- No natural ordering (e.g., transport mode: Bus, Car, Train).
- **Baseline Category Logit Model:** Choose category 1 as baseline.

$$\log\frac{\pi_j}{\pi_1} = \beta_{0j} + \boldsymbol{\beta}_j^T\mathbf{x}, \quad j = 2, \dots, J$$

- **Probability:** $\pi_1 = \frac{1}{1+\sum e^{\eta_k}}$, $\pi_j = \frac{e^{\eta_j}}{1+\sum e^{\eta_k}}$.
- **Goodness of Fit (Grouped):**
  - Deviance: $D = 2\sum_i\sum_j y_{ij}\log(y_{ij}/(n_i\hat{\pi}_{ij}))$
  - Pearson: $X^2 = \sum_i\sum_j \frac{(y_{ij} - n_i\hat{\pi}_{ij})^2}{n_i\hat{\pi}_{ij}}$
  - $df = (N - p)(J - 1)$ (N: groups, p: params per cat, J: categories).

2. Ordinal Response (Proportional Model)

- Natural ordering exists (e.g., Low, Medium, High).
- **Model Cumulative Probabilities:** $\gamma_j = P(Y \le j)$.

$$\text{logit}(P(Y \le j)) = \alpha_j - \boldsymbol{\beta}^T\mathbf{x}$$

- **Key Property:** The slope β is the same for all j. Only intercepts α_j differ.
- **Interpretation in R (`polr`):** $\beta > 0 \Rightarrow$ higher x **increases** the probability of higher categories (because $P(Y \le j)$ decreases).

IX. Exam Strategy & R Cheat Sheet

1. Interpreting `vcov(fit)`:

- Diagonal elements are $\text{Var}(\hat{\beta}_j)$.
- Off-diagonal are $\text{Cov}(\hat{\beta}_j, \hat{\beta}_k)$.
- $\text{Var}(A \pm B) = \text{Var}(A) + \text{Var}(B) \pm 2\text{Cov}(A, B)$.

2. Calculating Odds Ratios for Interactions:

- If model is $\eta = \beta_0 + \beta_{age}X_1 + \beta_{sex}X_2 + \beta_{int}X_1X_2$.
- Effect of age for males ($X_2 = 1$): $\exp(\beta_{age} + \beta_{int})$.